COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE-010: Data Structures and Algorithm	**SECTION:**	CPE21S3
DATE PERFORMED:	7/28/26	**DATE SUBMITTED:**	7/28/26
NAME:	Santos, Paterno Miguel R.	**INSTRUCTOR:**	Engr. Jimlord Quejado

Seatwork 4.1: Array-Based vs. STL Stack Implementations

1. PROCEDURE OUTPUT

What procedures did you perform?

ILO A:

CODE:

ILO_A > main.cpp > main()

```cpp
#include <iostream>
#include <stack>

int main(){

    std::stack<int> newStack; // Created object
    std::stack<int> swapStack; // For swapping

    // Testing the empty() function to check if stack is empty
    std::cout <<"Is the stack empty? "<< newStack.empty()<<std::endl;

    // Insert 10 elements in a for loop
    for(int i = 0; i < 10; i++){
        newStack.push(i);
        swapStack.push(i);

        // Prints out the first top
        std::cout <<"Top of the stack: "<< newStack.top()<<std::endl;
    }

    swapStack.swap(newStack);
    std::cout << "newStack new top value: " << newStack.top()<<std::endl;


    newStack.emplace(10);

    // Prints out the stack size
    std::cout <<"Size of the stack is: "<< newStack.size()<<std::endl;


    // Removing elements in a for loop
    while(!newStack.empty()){
        std::cout <<"Remaining elements of the stack is: "<< newStack.size()<<std::endl;
        newStack.pop();
    }
    std::cout << "No more elements remaining \n";
}
```

OUTPUT:

```
PS C:\Users\TIPQC\Documents\SANTOS\HOA4p1\ILO_A> cd "c:\Users\TIPQC\Documents\SANTOS\HOA4p1\ILO_A\" ; if ($?) { g++ main.cpp -o main } ; if ($?) { .\main }
Is the stack empty? 1
Top of the stack: 0
Top of the stack: 1
Top of the stack: 2
Top of the stack: 3
Top of the stack: 4
Top of the stack: 5
Top of the stack: 6
Top of the stack: 7
Top of the stack: 8
Top of the stack: 9
newStack new top value: 9
Size of the stack is: 11
Remaining elements of the stack is: 11
Remaining elements of the stack is: 10
Remaining elements of the stack is: 9
Remaining elements of the stack is: 8
Remaining elements of the stack is: 7
Remaining elements of the stack is: 6
Remaining elements of the stack is: 5
Remaining elements of the stack is: 4
Remaining elements of the stack is: 3
Remaining elements of the stack is: 2
Remaining elements of the stack is: 1
No more elements remaining
PS C:\Users\TIPQC\Documents\SANTOS\HOA4p1\ILO_A>
```

Code Analysis: I included a library for stack to access the commands to make stacks. I then declared to stacks with “newStack” being my main stack and “swapStack” for testing out the swapping commands. The empty command returns a boolean value to tell whether the stack is empty or not. Then I used a for loop to push 10 elements inside my main stack to insert 10 different values along with the other stack. Then I display the current top value of the main stack using the top command. Then I swapped the values of both stacks and used an emplace command to add a value of 10 at the top of the main stack. Afterwards, I checked the size of the stack and also deleted all the values of the main stack by using the pop command.

ILO B:

CODE:

ILO_B > main.cpp > displayAll()

```cpp
#include <iostream>

const size_t maxCap = 100;
int stack[maxCap];
int top = -1, newData, elements;

// Prototyping
void push();
void pop();
void Top();
bool isFull();
bool isEmpty();
void displayAll();

int main(){
    int choice;

    std:: cout << "Enter number of max element for new stack (max: 100)";
    std::cin >> elements;

    while(true){
        std::cout << "Stack operations: "<<std::endl;
        std::cout << "1. Push \n2. Pop \n3. Top \n4. isEmpty \n5. isFull \n6. Display All \n";
        std::cin >> choice;

        switch(choice){
            case 1: push();
            break;

            case 2: pop();
            break;

            case 3: Top();
            break;

            case 4: std::cout << isEmpty() << std::endl;
            break;

            case 5: std:: cout << isFull() << std::endl;
            break;

            case 6: displayAll();
            break;

            default: std::cout << "Invalid Choice" << std::endl;
            break;
        }
    }


```

```
    return 0;
}

bool isEmpty(){
    // Top is -1
    if(top == -1) return true;
    return false;
}

bool isFull(){
    // != maxCap
    if(top >= elements){
        std::cout << "Stack Overflow";
        return true;
    }
return false;
}

void push(){

    // Check if stack is full, if yes return error
    if(isFull())return;

    // Ask user for value to be pushed in stack
    std::cout<<"What number do you want to add to the stack? ";
    std::cin>> newData;

    // Store inside stack
    stack[++top] = newData;

}

void pop(){

    // Check if there are no more elements inside the stack
    if(isEmpty()) return;

    // Display element to be popped:
    std::cout<<"Popping: "<< stack[top];

    // Decrement top
    --top;
}

void Top(){
    // Error check
    if(isEmpty()) return;

    // Display top element
    std::cout << "The top element is: " << stack[top] << std::endl;
```

```
101 }
102
103 void displayAll(){
104     // Error check
105     if(isEmpty()) return;
106
107     std::cout << "Values inside the stack: \n";
108     for(int i = 0; i < elements; i++){
109         std::cout << stack[i] << std::endl;
110     }
111
112 }
113
```

OUTPUT:

```
PS C:\Users\TIPQC\Documents\SANTOS\HOA4p1\ILO_B> cd "c:\Users\TIPQC\Documents\SANTOS\HOA4p1\ILO_B\" ; if ($?) { g++ main.cpp -o main } ; if ($?) { .\main }
Enter number of max element for new stack (max: 100)100
Stack operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. Display All
1
What number do you want to add to the stack? 12
Stack operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. Display All
1
What number do you want to add to the stack? 23
Stack operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. Display All
1
What number do you want to add to the stack? 34
Stack operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. Display All
2
Popping: 34Stack operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. Display All
3
The top element is: 23
Stack operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. Display All
```

```
Popping: 34Stack operatic
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. Display All
3
The top element is: 23
Stack operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. Display All
4
0
Stack operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. Display All
5
0
Stack operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. Display All
6
Values inside the stack:
12
23
34
0
0
0
0
0
0
0
0
0
0
0
0
0
0
```

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Code Analysis: I declared a constant size of allowed bytes as the max capacity of 100 to limit the stack elements allowed along with three integers to use for later. I prototyped six functions and called the int main. I then made a menu for the command line interface for the functions. Using switch case, I called every function per case and a default as an invalid choice.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

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